

Unit 0.

**Introduction to
problem solving framework**

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AGENDA

Science

Inquiry/Research

Systems Thinking

Case Study Method for Learning

Science

Science as a body of knowledge
Science as what scientists do
Science as a social institution

Science as a whole

*System of science is
non separable*

Sociology of
science

Psychology
of science

What/where
is it ?

Logic of
science

Adapted from Mitroff (1974)

Science

Categories of science:

- ❑ natural science vs. **science of the artificial** (Simon 1969);
- ❑ natural vs. **design science** (March & Smith 1995);
- ❑ behavioral vs. **design science** (Hevner et al 2004);
- ❑ scientific vs. **engineering discipline** (Long & Dowell 1989); or
- ❑ explanatory vs. **design sciences** (van Aken 2004)

(truth) - building and justification of theories of description, explanation and prediction of real world phenomena

(utility) - design and implementation of artifacts for solving and improving real world problems

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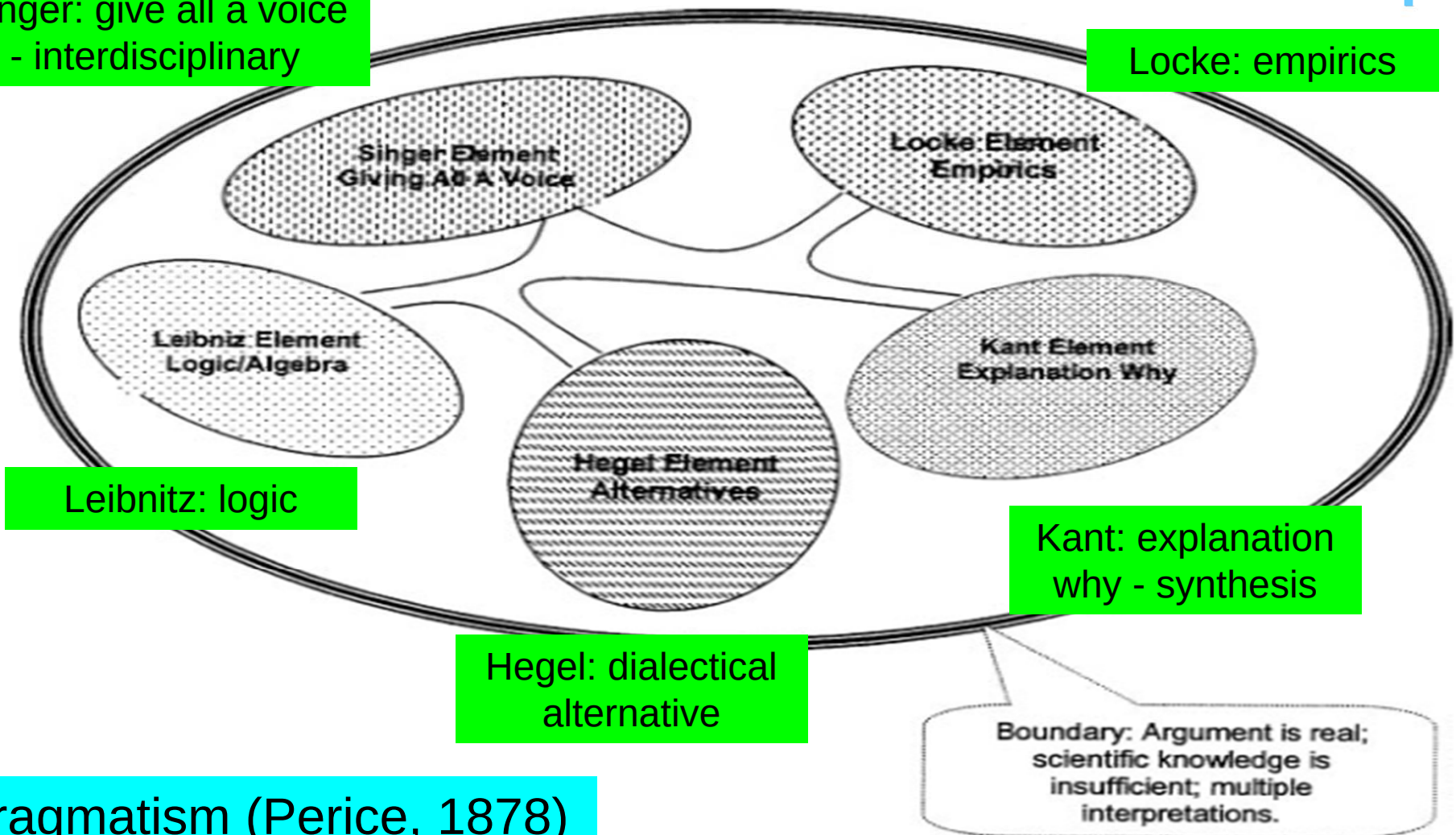
Empiricism, Rationalism, Criticism (Singer 1959)

Science*

Churchman's guarantors of truth

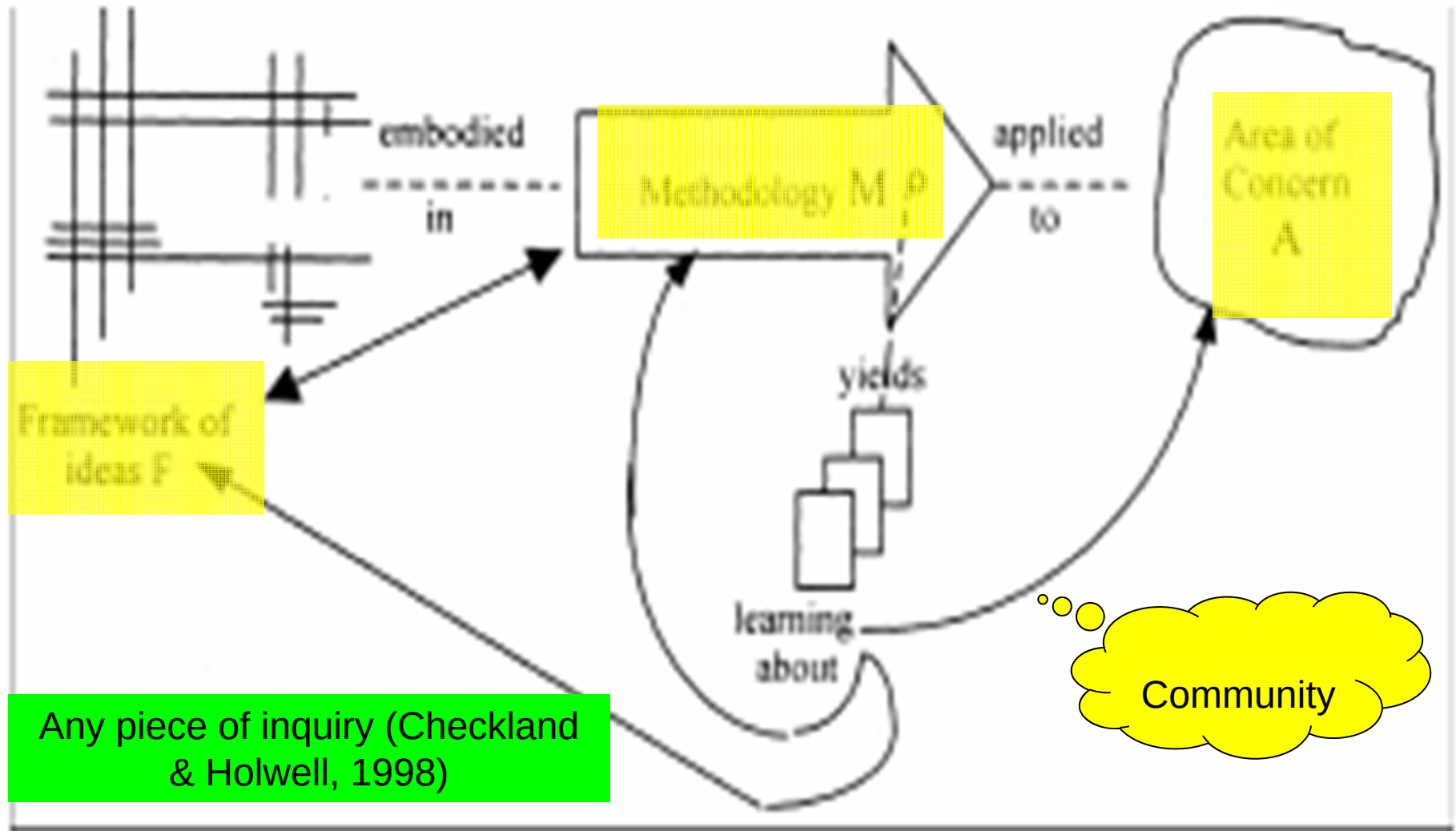
Singer: give all a voice
- interdisciplinary

Locke: empirics



Pragmatism (Perice, 1878)

Inquiry/Research



Inquiry/Research

British (Lockean) empiricism – induction

French (Descartian) rationalism - deduction

American (Peircean) pragmatism – abduction

Logic of justification

Logic of discovery

	deduction	induction	abduction
given	rule	case	rule
given	case	result/fact	result/fact
conclusion	result/fact	rule	case

Inquiry/Research

- DEDUCTION

- Rule.– All the beans from this bag are white.
- Case.– These beans are from this bag.
- Result.– These beans are white.

- INDUCTION

- Case.– These beans are from this bag.
- Result.– These beans are white.
- Rule.– All the beans from this bag are white.

- HYPOTHESIS

- Rule.– All the beans from this bag are white.
- Result.– These beans are white.
- Case.– These beans are from this bag.

SYSTEMS THINKING

Machine-age thinking	Systems-age thinking
industrial revolution, analysis, reductionism, cause-effect relations, determinism value free	post-industrial revolution, synthesis, expansionism, producer-product relations, free will and choice value full

Butterfly Effect

Chaos Theory

- ❑ Reaction to failures of **natural science** to complex, real-world problems of **social sciences** (Checkland 1981)

Systems thinking: "Synthesize, don't analyse!"

Ethical (Churchman 1971)

SYSTEMS THINKING

Scientific Thinking: only 1 system – 1 universe, 1 truth (Metcalf, 2006)

- ❑ **Concept list: boundary, purpose, stakeholder, process, contradictions,**
 - *Systems thinking vs. science thinking: boundary (Ackoff, 2000)*
- ❑ **Unit of analysis, mentally imagined, interconnected, bounded (Metcalf, 2006)**
- ❑ **Multiple Perspective:** To be ethical, argument should give all stakeholders a voice and encourage innovative rationally justified ways of seeing situations (Churchman, Mason, Mitroff, Linstone)

**Linstone (1984, 1993) TOP perspectives
(technical, organizational, personal)**

SYSTEMS THINKING

- Analysis is a three-step thought process
 - First, it takes apart that which it seeks to understand.
 - Then it attempts to explain the behavior of the parts taken separately.
 - Finally, it tries to aggregate understanding of the parts into an explanation of the whole.
- Systems thinking puts the system in the context of the larger environment of which it is a part and studies the role it plays in the larger whole

Gharajedaghi, 2011

CASE STUDY METHOD FOR LEARNING (Ellet 2007)

- In the *lecture* method, learners receive knowledge from an expert.
- In the *case* method, learners make the knowledge with the assistance of an expert.
 - This fundamental shift causes many new case method students to be confused and uncertain about how they should go about learning.
- *Persuasion through argument*

CASE STUDY METHOD FOR LEARNING (Ellet 2007)

What? (conclusion)

The president is right to default on the country's foreign debt.

[because]

Why? (reason)

Full payment of the debt will destabilize the country.

Evidence

1. Debt payments will take money out of the country that is badly needed to support the economy and meet social needs such as education and healthcare.

Evidence includes historical narrative showing that past economic downturns have impoverished the population and created political conflict that worsened the economic situation.

2. National finances are depleted.

Evidence includes data and calculations showing that poor management of the economy has resulted in deficit spending and spotty tax collection.

3. The national economy is just beginning to recover.

CASE STUDY METHOD FOR LEARNING (Ellet 2007)

- Working on cases:

C1. Situation

C2. Questions

C3. Hypothesis

C4. Proof and action

C5. Alternatives

CASE STUDY METHOD FOR LEARNING (Ellet 2007)

C1. Situation (5 minutes)

- Understanding the big picture first and then filling it in with details.
- Start by asking this question: ***What is the situation?***
- Usually reading the ***first*** and ***last*** sections of the case is sufficient to identify the situation

CASE STUDY METHOD FOR LEARNING (Ellet 2007)

C2. Questions (15 minutes)

- The most important = *What do I need to know about the situation?*

C21. [PROBLEM BASED QUESTIONS]

- Who or what is the subject of the problem (e.g., a manager, a company, a country)?
- What is the problem? Am I trying to account for a failure, a success, or something more ambiguous?
- What's the significance of the problem to the subject?
- Who is responsible for the problem (usually it is the protagonist) and what might he need to know to do something about it?

CASE STUDY METHOD FOR LEARNING (Ellet 2007)

C3. Hypothesis (45 minutes)

- *The most important phase of work on the case*
- Through close study of high value sections and exhibits, you narrow the possibilities to the one *that seems most plausible to you*
- My hypothesis is wrong?
 - *A hypothesis isn't wrong*; a hypothesis fails when you can't make a credible argument for it from case evidence

CASE STUDY METHOD FOR LEARNING (Ellet 2007)

C31. [PROBLEM BASED HYPOTHESES]

- Make sure you know the problem that needs to be diagnosed. Consider whether the characteristics of the problem suggest causes.
- Think about the frameworks that seem most appropriate to the situation. Quickly review the specifics of the frameworks if you aren't sure of them.
- Pursue the diagnosis by looking at case information through the lens of the cause you are most certain about.
- For each cause, make a separate pass through the case looking for evidence of it.
- If the case has a lot of quantitative evidence, to what cause is it most relevant? If you don't have a cause relevant to the quantitative evidence, formulate one. Work up as much relevant, high-value quantitative evidence as you can.
- In a case with a protagonist, consider whether she is a potential cause. If you think she is, work out how she contributes to the problem.

CASE STUDY METHOD FOR LEARNING (Ellet 2007)

C4. Proof and Action (40 minutes)

- You want to prove something, not look for something to prove.
- **Ask these questions:**
 - What evidence do I have that supports the hypothesis?
 - What additional evidence do I need?
- Go back into the case, with the single purpose of bringing out more evidence that aligns with your hypothesis.
- You don't have to work from the first page to the last. You can go directly to the sections and exhibits you think have what you need.
- Of course, you can work from beginning to end if that makes you more comfortable. Just be sure to stay focused on what you're trying to prove

CASE STUDY METHOD FOR LEARNING (Ellet 2007)

C5. Alternatives (15 minutes)

- The last phase of analysis should be *to question your own hypothesis*:
 - What is the greatest weakness of the hypothesis?
What is the strongest alternative to it?

C51. [PROBLEM BASED ALTERNATIVES]

- Can the problem be defined differently? Would that make a difference to the diagnosis?
- Are there any holes in the diagnosis—could there be causes missing?
- What's the weakest part of the diagnosis?
- Could an entirely different diagnosis be made?
What would it look like?



THE END

